

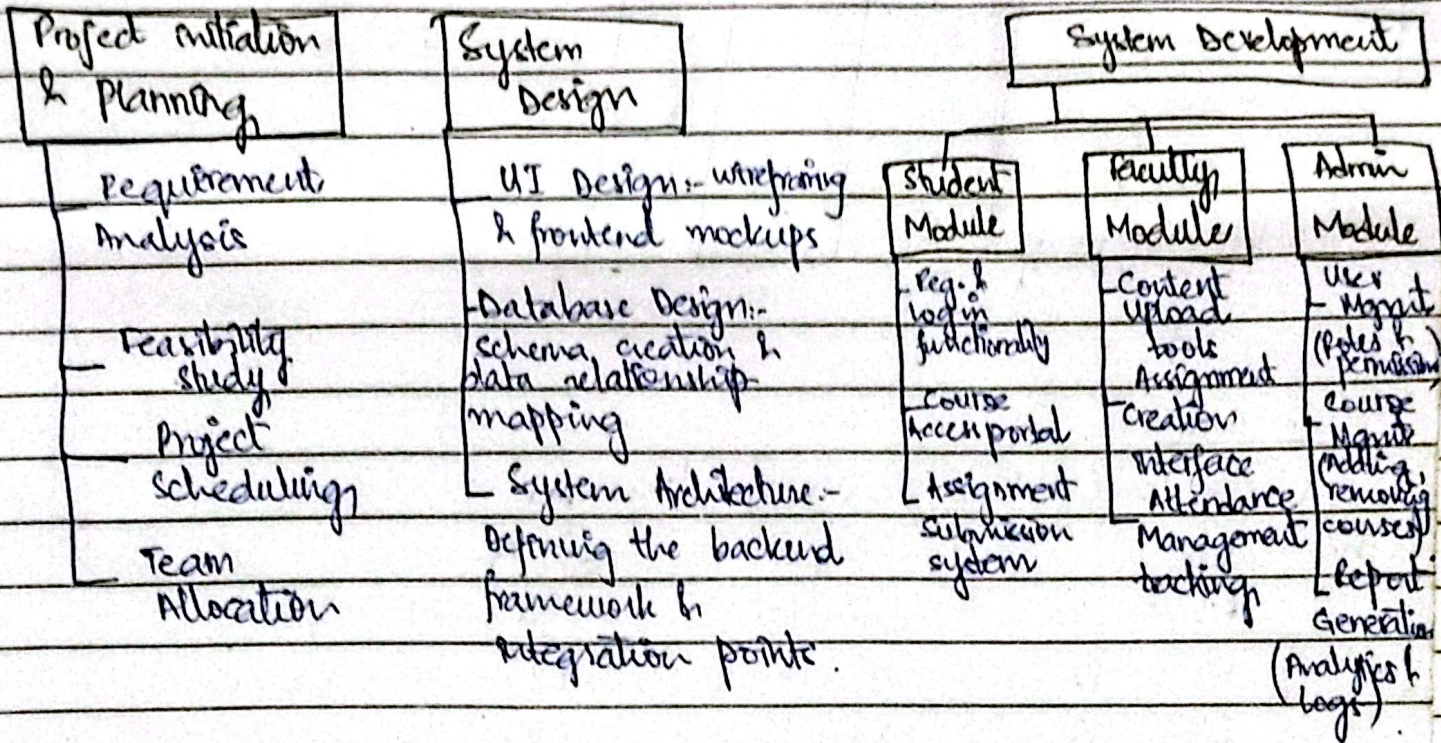
- hierarchical decomposition of the total scope of work
- to be carried out by the project team.
- create the required deliverables.

Date: _____

M	T	W	T	F	S	S
---	---	---	---	---	---	---

WBS → Work Breakdown Structure

for iLearn :-



CASE STUDY Topic :- Project Estimation
chat Application:-

Member: Salas	R1	R2	R3	Member: Aliya	R1	R2	R3
Create UI	3	3	3	Create UI	4	4	4
Create Groups	3	3	4	Create Groups	6	5	5
Send message	6	6	6	Send message	2	5	5
Receive message	3	3	3	Receive message	3	3	3
Add member in groups	5	2	2	Add member in groups	1	1	2
$\Sigma = 20$	17	18		$\Sigma = 16$	18	19	

Member: Ah	R1	R2	R3	R1: Round 1
Create UI	7	7	7	R2: Round 2
Create Groups	8	3	3	R3: Round 3
Send message	7	6	5	
Receive message	2	2	3	
Add member in groups	5	2	2	
	$\Sigma = 29$	20	20	

Result Graph :-

↳ we have three estimators / Members

① Salas

R1 :- 20

R2 :- 17

R3 :- 18

② Aya

R1 :- 16

R2 :- 18

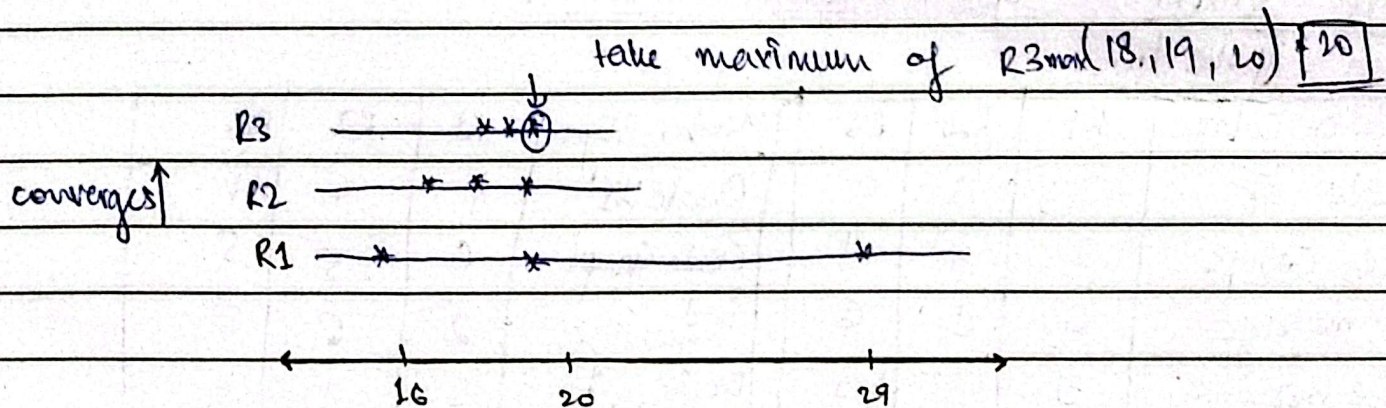
R3 :- 19

③ Ah

R1 :- 29

R2 :- 20

R3 :- 20



Estimation chart for R1 :-

R1		E1	E2	E3	Best case	Worst case	Average Time = $\frac{B+W}{2}$
Tasks →	Create UI	3	4	7	7	3	5
	Create Groups	3	6	8	8	3	$[5.5] = 6$
	Send message	6	2	7	7	2	$[4.5] = 5$
	Receive message	3	3	2	2	3	$[2.5] = 3$
	Add member in groups	5	5	5	1	5	3

Page No. _____

WES®

Signature _____

Date: _____

M	T	W	T	F	S	S
---	---	---	---	---	---	---

Estimation chart for R2:-

R2.		E1	E2	E3	Best	Worst	Average
	Create UI	3	4	7	3	7	5
	Create Groups	3	5	3	3	5	4
	Send message	6	5	6	5	6	6
	Receive message	3	3	2	2	3	3
	Add member in groups	2	1	2	1	3	2

Estimation chart for R3:-

R3.		E1	E2	E3	Best	Worst	Average
	Create UI	3	4	7	3	7	5
	Create Groups	4	5	3	3	5	4
	Send message	6	5	5	5	6	6
	Receive message	3	3	3	3	3	3
	Add member in groups	2	2	2	2	2	2

Delta Table:-

	$\Delta 1$	$\Delta 2$	$\Delta 3$
Create UI	4	4	4
Create Groups	5	2	2
Send message	5	1	1
Receive message	1	1	0
Add member in Groups	4	2	0

Date: _____

M	T	W	T	F	S	S
---	---	---	---	---	---	---

WBS (Work Breakdown Structure) :-

↳ When to develop WBS?

After finalizing project scope & before project estimation.

Levels in WBS :-

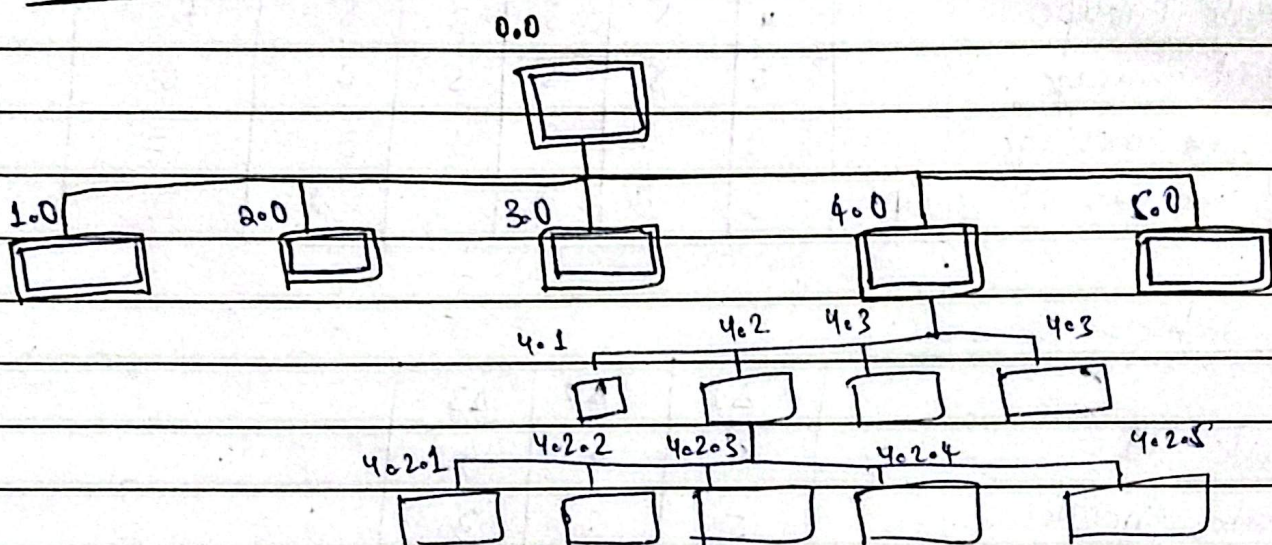
↳ Start with level 0, i.e. project title.

↳ Level 1 → highest level breakdown of about 4-7 nodes depending on project. Termed as control accounts.

↳ Level 2 → work packages

↳ Level 3 → tasks / activities to be performed.

Graphical Free WBS



Make sure :-

Each level has no more than 7 items.

WIDE BAND DELPHI Process :-

1. Choose the team → team 3 to 7 members along with a moderator

2. Kick-off meeting → team brainstorm & write down assumptions

↳ team has generated WBS, & team agrees on a unit of estimation.

Page No. _____

WBS®

(hr/day) Signature _____

-- PM :- Project Manager

Date: _____

M	T	W	T	F	S	S
---	---	---	---	---	---	---

3. Individual preparation

Every team member writes down estimations & assumptions.

4. Estimation Session :- multiple rounds until converge

5. Assemble Tasks :- ^{PM} Compiles the final list, estimates & assumptions

6. Review Results :- PM reviews the final task list with the estimation team.

CHAPT 4 - REQUIREMENTS ENGINEERING

Requirements two types :-

-- User requirements → natural lang^o, non technical terms, client, end-user

-- System requirements → structured doc^o, system's fn^s, services & constraints, scope.
↳ developer, testers etc

-- FR

-- NPR

↳ product requirements

These specify how the product/system should behave in terms of performance & quality.

Ex: The system shall respond within 2 seconds.

↳ The system shall support 10,000 concurrent users.

↳ The system shall have 99.9% availability.

-- Performance, Reliability, Usability, Security, Efficiency, Portability
↳ these are measurable & testable.

↳ Organizational Requirements

↳ internal policies, standards or constraints

↳ The system shall follow Agile Methodology.

↳ The system documentation shall follow IEEE format.

Date: _____

M	T	W	T	F	S	S
---	---	---	---	---	---	---

↳ External Requirements

External factors → legal ; regulatory or industry standard.

↳ The system shall comply with data protection laws

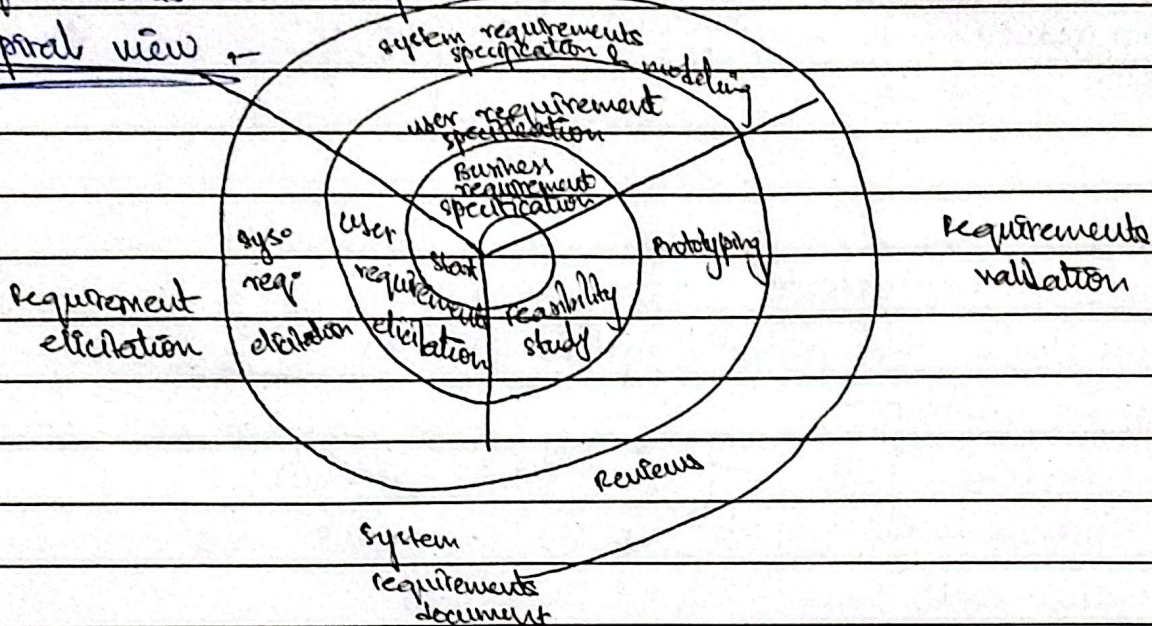
↳ The system shall generate tax reports according to the govt rules.

Requirements Imprecision :-

↳ less details → ambiguous requirements → system delay , increased costs

Requirements → complete & consistent.

Spiral view



Requirement Discovery :-

↳ Interviewing

↳ Closed interview :- pre-determined list of Qs

↳ Open interview :- no specific questionnaire, explore req., & refine them.

↳ Ethnography.

